

- 7 (a) State what is meant by the root-mean-square (r.m.s.) value of an alternating current.

.....

.....

..... [2]

- (b) There is a sinusoidal alternating current in a resistor of resistance $26\ \Omega$. The power P dissipated in the resistor varies with time t as shown in Figure 7.1.

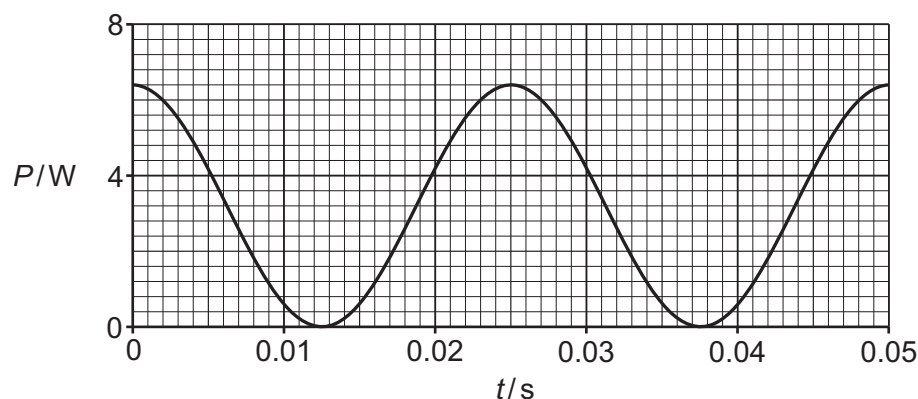


Figure 7.1

- (i) Determine the mean power dissipated in the resistor.

mean power = W [1]

- (ii) Calculate the r.m.s. current in the resistor.

r.m.s. current = A [2]

(c) The current in the resistor in **7(b)** is now half-wave rectified.

(i) State the name of the electric circuit component that is used to achieve this rectification.

..... [1]

(ii) On Figure 7.2, sketch the variation of the power P dissipated in the resistor between $t = 0$ and $t = 0.050$ s.

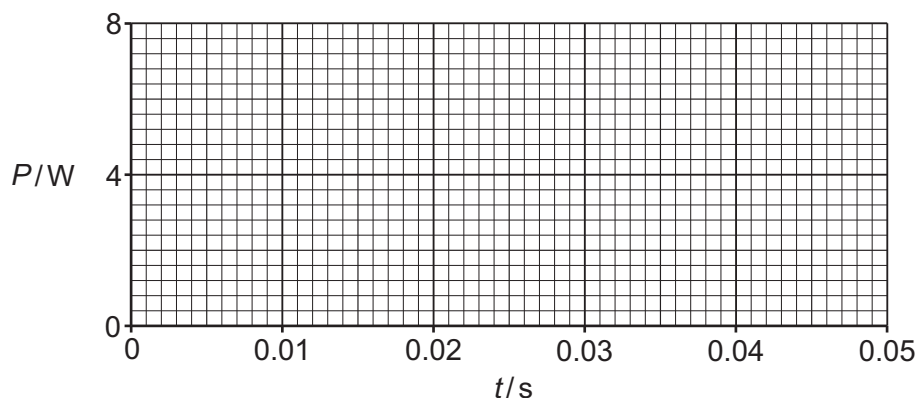


Figure 7.2

[2]

(iii) Determine the r.m.s. current in the resistor. Explain your reasoning.

r.m.s. current = A [2]

[Total: 10]